

Time Series — Solutions

Practice Exam — Week 4: Parametric Estimation — Yule–Walker & Least Squares
EPFL, MATH-342 · Prof. S. C. Olhede

Problem 1 (5 points)

Statement.

Let $\{X_t\}$ be a mean-zero causal AR(2) process $X_t = \phi_1 X_{t-1} + \phi_2 X_{t-2} + \varepsilon_t$.

(a) Starting from the model equation, derive the Yule–Walker equations: multiply by X_{t-k} , take expectations, and explain precisely where causality is used to kill the noise cross-term. State the resulting linear system $\gamma = \Gamma\phi$ and the variance identity for σ^2 . (2 pts)

(b) A series gives the sample autocovariances $\hat{\gamma}_0 = 5$, $\hat{\gamma}_1 = 3$, $\hat{\gamma}_2 = 1$. Compute the Yule–Walker estimates $\hat{\phi}_1$, $\hat{\phi}_2$ and $\hat{\sigma}^2$. (2 pts)

(c) Verify that the fitted model of part (b) is causal (stationary) by locating the roots of its AR polynomial. (1 pt)

Solution.

(a) *Derivation.* Multiply the model $X_t = \phi_1 X_{t-1} + \phi_2 X_{t-2} + \varepsilon_t$ by X_{t-k} and take expectations. With mean zero, $\mathbb{E}[X_{t-j}X_{t-k}] = \gamma_{k-j}$, so for every $k \geq 0$,

$$\gamma_k = \phi_1 \gamma_{k-1} + \phi_2 \gamma_{k-2} + \mathbb{E}[\varepsilon_t X_{t-k}].$$

Because $\{X_t\}$ is *causal*, X_{t-k} is a one-sided combination of $\varepsilon_{t-k}, \varepsilon_{t-k-1}, \dots$ (innovations *strictly earlier* than ε_t when $k \geq 1$), and white noise is uncorrelated across time, so $\mathbb{E}[\varepsilon_t X_{t-k}] = 0$ for $k \geq 1$. This is the single place where causality enters. Taking $k = 1, 2$ gives, with $\gamma_{-1} = \gamma_1$,

$$\gamma = \Gamma\phi, \quad \begin{pmatrix} \gamma_1 \\ \gamma_2 \end{pmatrix} = \begin{pmatrix} \gamma_0 & \gamma_1 \\ \gamma_1 & \gamma_0 \end{pmatrix} \begin{pmatrix} \phi_1 \\ \phi_2 \end{pmatrix}.$$

For $k = 0$ the cross-term does *not* vanish: multiplying instead by X_t , $\mathbb{E}[\varepsilon_t X_t] = \mathbb{E}[\varepsilon_t(\phi_1 X_{t-1} + \phi_2 X_{t-2} + \varepsilon_t)] = \sigma^2$ (all $\mathbb{E}[\varepsilon_t X_{t-j}] = 0$ for $j \geq 1$), giving the variance identity

$$\gamma_0 = \phi_1 \gamma_1 + \phi_2 \gamma_2 + \sigma^2, \quad \text{i.e.} \quad \sigma^2 = \gamma_0 - \phi_1 \gamma_1 - \phi_2 \gamma_2.$$

(b) With $\hat{\gamma}_0 = 5$, $\hat{\gamma}_1 = 3$, $\hat{\gamma}_2 = 1$ the Yule–Walker system is

$$\begin{pmatrix} 5 & 3 \\ 3 & 5 \end{pmatrix} \begin{pmatrix} \hat{\phi}_1 \\ \hat{\phi}_2 \end{pmatrix} = \begin{pmatrix} 3 \\ 1 \end{pmatrix}.$$

The inverse of $\begin{pmatrix} a & b \\ b & a \end{pmatrix}$ is $\frac{1}{a^2 - b^2} \begin{pmatrix} a & -b \\ -b & a \end{pmatrix}$; here $a = 5$, $b = 3$, determinant $a^2 - b^2 = 25 - 9 = 16$, so

$$\hat{\phi} = \frac{1}{16} \begin{pmatrix} 5 & -3 \\ -3 & 5 \end{pmatrix} \begin{pmatrix} 3 \\ 1 \end{pmatrix} = \frac{1}{16} \begin{pmatrix} 15 - 3 \\ -9 + 5 \end{pmatrix} = \frac{1}{16} \begin{pmatrix} 12 \\ -4 \end{pmatrix} = \begin{pmatrix} \frac{3}{4} \\ -\frac{1}{4} \end{pmatrix}.$$

So $\hat{\phi}_1 = 0.75$, $\hat{\phi}_2 = -0.25$. The variance estimate is

$$\hat{\sigma}^2 = \hat{\gamma}_0 - \hat{\phi}_1 \hat{\gamma}_1 - \hat{\phi}_2 \hat{\gamma}_2 = 5 - \frac{3}{4}(3) - \left(-\frac{1}{4}\right)(1) = 5 - \frac{9}{4} + \frac{1}{4} = 5 - 2 = 3.$$

Hence $\hat{\phi}_1 = \frac{3}{4}$, $\hat{\phi}_2 = -\frac{1}{4}$, $\hat{\sigma}^2 = 3$.

(c) The fitted AR polynomial is $\Phi(z) = 1 - \frac{3}{4}z + \frac{1}{4}z^2$. Solving $\frac{1}{4}z^2 - \frac{3}{4}z + 1 = 0$, i.e. $z^2 - 3z + 4 = 0$,

$$z = \frac{3 \pm \sqrt{9 - 16}}{2} = \frac{3 \pm \sqrt{-7}}{2} = \frac{3}{2} \pm \frac{\sqrt{7}}{2}i, \quad |z| = \sqrt{\left(\frac{3}{2}\right)^2 + \left(\frac{\sqrt{7}}{2}\right)^2} = \sqrt{\frac{9}{4} + \frac{7}{4}} = \sqrt{4} = 2.$$

Both roots have modulus $2 > 1$, i.e. lie outside the unit circle, so the fit is **causal (stationary)**. (Equivalently, the AR(2) triangle: $\phi_1 + \phi_2 = \frac{1}{2} < 1$, $\phi_2 - \phi_1 = -1 < 1$, $|\phi_2| = \frac{1}{4} < 1$.)

Problem 2 (5 points)

Statement.

A series of length $n = 150$ is modelled as a mean-zero AR(3) process $X_t = \phi_1 X_{t-1} + \phi_2 X_{t-2} + \phi_3 X_{t-3} + \varepsilon_t$. The estimated autocorrelations are

$$\hat{\rho}_1 = 0.8, \quad \hat{\rho}_2 = 0.6, \quad \hat{\rho}_3 = 0.4,$$

and the sample variance is $\hat{\gamma}_0 = 10$.

(a) Explain why the Yule–Walker fit can be written entirely with the correlation matrix, $\hat{\phi} = \bar{\Gamma}_3^{-1} \hat{\rho}_3$, i.e. why $\hat{\gamma}_0$ cancels in the system for ϕ . Write out the 3×3 system explicitly. (1.5 pts)

(b) Solve the system for $\hat{\phi}_1, \hat{\phi}_2, \hat{\phi}_3$ (e.g. by Gaussian elimination) and verify your solution by substituting it back into the three equations. (2.5 pts)

(c) Compute the innovation-variance estimate $\hat{\sigma}^2$ using $\hat{\gamma}_0 = 10$. (1 pt)

Solution.

(a) The population Yule–Walker system is $\gamma = \Gamma \phi$, with $\Gamma = (\gamma_{i-j})$ and $\gamma = (\gamma_1, \gamma_2, \gamma_3)^\top$. Factor out γ_0 : writing $V = \gamma_0^{-1} I$ and multiplying both sides by V ,

$$V\gamma = V\Gamma\phi \implies \hat{\rho}_3 = \bar{\Gamma}_3 \phi,$$

where every entry has been divided by γ_0 , so $\bar{\Gamma}_3 = (\hat{\rho}_{i-j})$ and $\hat{\rho}_3 = (\hat{\rho}_1, \hat{\rho}_2, \hat{\rho}_3)^\top$. The factor γ_0 cancels because it multiplies *both* sides equally; hence $\hat{\phi} = \bar{\Gamma}_3^{-1} \hat{\rho}_3$ depends only on the correlations. With $\hat{\rho}_0 = 1$ the explicit system is

$$\begin{pmatrix} 1 & 0.8 & 0.6 \\ 0.8 & 1 & 0.8 \\ 0.6 & 0.8 & 1 \end{pmatrix} \begin{pmatrix} \hat{\phi}_1 \\ \hat{\phi}_2 \\ \hat{\phi}_3 \end{pmatrix} = \begin{pmatrix} 0.8 \\ 0.6 \\ 0.4 \end{pmatrix}.$$

(b) Solve by elimination. Label the rows R_1, R_2, R_3 .

$$R_1 : \quad \hat{\phi}_1 + 0.8\hat{\phi}_2 + 0.6\hat{\phi}_3 = 0.8,$$

$$R_2 : \quad 0.8\hat{\phi}_1 + \hat{\phi}_2 + 0.8\hat{\phi}_3 = 0.6,$$

$$R_3 : \quad 0.6\hat{\phi}_1 + 0.8\hat{\phi}_2 + \hat{\phi}_3 = 0.4.$$

Eliminate $\hat{\phi}_1$. Compute $R_2 - 0.8R_1$ and $R_3 - 0.6R_1$:

$$R'_2 : \quad (1 - 0.64)\hat{\phi}_2 + (0.8 - 0.48)\hat{\phi}_3 = 0.6 - 0.64 \implies 0.36\hat{\phi}_2 + 0.32\hat{\phi}_3 = -0.04,$$

$$R'_3 : \quad (0.8 - 0.48)\hat{\phi}_2 + (1 - 0.36)\hat{\phi}_3 = 0.4 - 0.48 \implies 0.32\hat{\phi}_2 + 0.64\hat{\phi}_3 = -0.08.$$

From R'_3 , divide by 0.32: $\hat{\phi}_2 + 2\hat{\phi}_3 = -0.25$. From R'_2 , divide by 0.04: $9\hat{\phi}_2 + 8\hat{\phi}_3 = -1$. Substitute $\hat{\phi}_2 = -0.25 - 2\hat{\phi}_3$:

$$9(-0.25 - 2\hat{\phi}_3) + 8\hat{\phi}_3 = -1 \implies -2.25 - 10\hat{\phi}_3 = -1 \implies \hat{\phi}_3 = \frac{-1.25}{10} = -\frac{1}{8} = -0.125.$$

Then $\hat{\phi}_2 = -0.25 - 2(-0.125) = -0.25 + 0.25 = 0$, and from R_1 , $\hat{\phi}_1 = 0.8 - 0.8(0) - 0.6(-0.125) = 0.8 + 0.075 = 0.875 = \frac{7}{8}$. Thus

$$\boxed{\hat{\phi}_1 = \frac{7}{8} = 0.875, \quad \hat{\phi}_2 = 0, \quad \hat{\phi}_3 = -\frac{1}{8} = -0.125}$$

Check. $R_1 : 0.875 + 0.8(0) + 0.6(-0.125) = 0.875 - 0.075 = 0.8$. $\checkmark$ $R_2 : 0.8(0.875) + 0 + 0.8(-0.125) = 0.7 - 0.1 = 0.6$. $\checkmark$ $R_3 : 0.6(0.875) + 0.8(0) + (-0.125) = 0.525 - 0.125 = 0.4$. $\checkmark$

(c) Using $\hat{\sigma}^2 = \hat{\gamma}_0(1 - \sum_j \hat{\phi}_j \hat{\rho}_j)$ with $\hat{\gamma}_0 = 10$,

$$1 - \hat{\phi}_1 \hat{\rho}_1 - \hat{\phi}_2 \hat{\rho}_2 - \hat{\phi}_3 \hat{\rho}_3 = 1 - \frac{7}{8}(0.8) - 0(0.6) - \left(-\frac{1}{8}\right)(0.4) = 1 - 0.7 + 0.05 = 0.35 = \frac{7}{20},$$

hence $\hat{\sigma}^2 = 10 \cdot \frac{7}{20} = \frac{7}{2} = 3.5$. (The series length $n = 150$ plays no role in the point estimates; it would only enter standard errors.)

Problem 3 (5 points)

Statement.

This problem treats the two faces of least squares.

(a) (MA via residual reconstruction.) A mean-zero MA(1) process $X_t = \varepsilon_t - \theta\varepsilon_{t-1}$ is observed at $X_1 = 0$, $X_2 = 2$, $X_3 = -1$. Reconstruct the innovations $\hat{\varepsilon}_1, \hat{\varepsilon}_2, \hat{\varepsilon}_3$ as functions of θ (state your initialisation), write the least-squares criterion $S(\theta)$, and find $\hat{\theta}$. Is the fit invertible? (2.5 pts)

(b) (Forward least squares for AR(2).) For the five observations $x_1, \dots, x_5 = 1, 2, 0, -1, -2$ (already mean-zero), write the forward design $\mathbf{X}_F = F\boldsymbol{\phi} + \boldsymbol{\varepsilon}$ for an AR(2) fit, form the normal equations $F^\top F \hat{\boldsymbol{\phi}}_F = F^\top \mathbf{X}_F$, and solve for $\hat{\boldsymbol{\phi}}_F$. Then report $\hat{\sigma}_F^2$ using the correct degrees of freedom. (2 pts)

(c) State, in one sentence each, (i) why the backward least-squares estimator $\hat{\boldsymbol{\phi}}_B$ estimates the same coefficients $\boldsymbol{\phi}$, and (ii) whether a least-squares AR fit is guaranteed to be stationary. (0.5 pt)

Solution.

(a) The innovations are latent, so reconstruct them from the model equation $\varepsilon_t = X_t + \theta\varepsilon_{t-1}$. Initialise the pre-sample innovation to zero, $\varepsilon_0 = 0$ (take $t = 0$ as the start). Then

$$\hat{\varepsilon}_1 = X_1 + \theta\varepsilon_0 = 0, \quad \hat{\varepsilon}_2 = X_2 + \theta\hat{\varepsilon}_1 = 2, \quad \hat{\varepsilon}_3 = X_3 + \theta\hat{\varepsilon}_2 = -1 + 2\theta.$$

The least-squares criterion is

$$S(\theta) = \sum_{t=1}^3 \hat{\varepsilon}_t^2 = 0^2 + 2^2 + (2\theta - 1)^2 = 4 + (2\theta - 1)^2.$$

Only the last term depends on θ (because $X_1 = 0$ froze $\hat{\varepsilon}_1 = 0, \hat{\varepsilon}_2 = 2$), and it is minimised when $2\theta - 1 = 0$. Hence

$$\boxed{\hat{\theta} = \frac{1}{2}} \quad \text{with } S(\hat{\theta}) = 4.$$

Since $|\hat{\theta}| = \frac{1}{2} < 1$ (equivalently $\Theta(z) = 1 - \frac{1}{2}z$ has root $z = 2$ outside the unit circle), the fit is **invertible**.

(b) For AR(2) ($p = 2$) the forward regression uses the observable times $t = 3, 4, 5$, regressing each x_t on (x_{t-1}, x_{t-2}) :

$$\mathbf{X}_F = \begin{pmatrix} x_3 \\ x_4 \\ x_5 \end{pmatrix} = \begin{pmatrix} 0 \\ -1 \\ -2 \end{pmatrix}, \quad F = \begin{pmatrix} x_2 & x_1 \\ x_3 & x_2 \\ x_4 & x_3 \end{pmatrix} = \begin{pmatrix} 2 & 1 \\ 0 & 2 \\ -1 & 0 \end{pmatrix}.$$

Form the normal equations $F^\top F \hat{\phi}_F = F^\top \mathbf{X}_F$:

$$F^\top F = \begin{pmatrix} 2 & 0 & -1 \\ 1 & 2 & 0 \end{pmatrix} \begin{pmatrix} 2 & 1 \\ 0 & 2 \\ -1 & 0 \end{pmatrix} = \begin{pmatrix} 5 & 2 \\ 2 & 5 \end{pmatrix}, \quad F^\top \mathbf{X}_F = \begin{pmatrix} 2 & 0 & -1 \\ 1 & 2 & 0 \end{pmatrix} \begin{pmatrix} 0 \\ -1 \\ -2 \end{pmatrix} = \begin{pmatrix} 2 \\ -2 \end{pmatrix}.$$

With $\det(F^\top F) = 25 - 4 = 21$,

$$\hat{\phi}_F = \frac{1}{21} \begin{pmatrix} 5 & -2 \\ -2 & 5 \end{pmatrix} \begin{pmatrix} 2 \\ -2 \end{pmatrix} = \frac{1}{21} \begin{pmatrix} 10 + 4 \\ -4 - 10 \end{pmatrix} = \frac{1}{21} \begin{pmatrix} 14 \\ -14 \end{pmatrix} = \begin{pmatrix} \frac{2}{3} \\ -\frac{2}{3} \end{pmatrix}.$$

Residuals $r_t = x_t - \hat{\phi}_1 x_{t-1} - \hat{\phi}_2 x_{t-2}$ for $t = 3, 4, 5$:

$$\begin{aligned} r_3 &= 0 - \frac{2}{3}(2) + \frac{2}{3}(1) = -\frac{2}{3}, \\ r_4 &= -1 - \frac{2}{3}(0) + \frac{2}{3}(2) = -1 + \frac{4}{3} = \frac{1}{3}, \\ r_5 &= -2 - \frac{2}{3}(-1) + \frac{2}{3}(0) = -2 + \frac{2}{3} = -\frac{4}{3}, \end{aligned} \quad \left\| \mathbf{X}_F - F \hat{\phi}_F \right\|^2 = \frac{4}{9} + \frac{1}{9} + \frac{16}{9} = \frac{21}{9} = \frac{7}{3}.$$

The degrees of freedom are $n - 2p = 5 - 4 = 1$, so

$$\hat{\sigma}_F^2 = \frac{\left\| \mathbf{X}_F - F \hat{\phi}_F \right\|^2}{n - 2p} = \frac{7/3}{1} = \frac{7}{3} \approx 2.33.$$

So $\hat{\phi}_F = (\frac{2}{3}, -\frac{2}{3})^\top$, $\hat{\sigma}_F^2 = \frac{7}{3}$.

(c) (i) A stationary Gaussian AR(p) run *backwards* in time is the *same* AR(p) model (its ACVS is symmetric, $\gamma_{-\tau} = \gamma_\tau$, so forward and reversed processes have identical second-order structure), so regressing each point on its *future* lags estimates the very same coefficients ϕ . (ii) **No**: the OLS objective only minimises the residual sum of squares and places no constraint on the root moduli, so $\hat{\Phi}(z)$ may have roots inside the unit circle. (Here it happens to be stationary — $\Phi(z) = 1 - \frac{2}{3}z + \frac{2}{3}z^2$ has roots $z = \frac{1 \pm \sqrt{-5}}{2}$ of modulus $\sqrt{3/2} \approx 1.22 > 1$ — but this is not guaranteed in general; Yule–Walker, by contrast, always yields a stationary fit.)

Problem 4 (5 points)

Statement.

(Revision of Week 3.) Consider the ARMA(2, 1) process

$$\left(1 - \frac{3}{4}B + \frac{1}{8}B^2\right)X_t = \left(1 + \frac{1}{3}B\right)\varepsilon_t.$$

(a) Factor the AR polynomial $\Phi(z) = 1 - \frac{3}{4}z + \frac{1}{8}z^2$, locate its roots, and confirm the process is causal. Then check invertibility from $\Theta(z) = 1 + \frac{1}{3}z$. (2 pts)

(b) Using a partial-fraction expansion of $G(z) = \Theta(z)/\Phi(z)$, find a closed form for the MA(∞) weights ψ_j in $X_t = \sum_{j \geq 0} \psi_j \varepsilon_{t-j}$, and report ψ_0, ψ_1, ψ_2 . (3 pts)

Solution.

(a) The AR polynomial is $\Phi(z) = 1 - \frac{3}{4}z + \frac{1}{8}z^2$. Multiplying by 8, $z^2 - 6z + 8 = (z - 2)(z - 4)$, so

$$\Phi(z) = \frac{1}{8}(z - 2)(z - 4) = \left(1 - \frac{1}{2}z\right)\left(1 - \frac{1}{4}z\right),$$

with roots $z = 2$ and $z = 4$. Both satisfy $|z| > 1$, i.e. lie outside the unit circle, so the process is **causal** (stationary). The MA polynomial $\Theta(z) = 1 + \frac{1}{3}z$ has root $z = -3$, $|-3| = 3 > 1$ outside

the unit circle, so the process is also **invertible** (this guarantees $G = \Theta/\Phi$ is well defined on $|z| \leq 1$).

(b) Write $X_t = G(B)\varepsilon_t$ with

$$G(z) = \frac{1 + \frac{1}{3}z}{(1 - \frac{1}{2}z)(1 - \frac{1}{4}z)}.$$

Partial-fraction in the reciprocal-root form

$$G(z) = \frac{A}{1 - \frac{1}{2}z} + \frac{B}{1 - \frac{1}{4}z}, \quad 1 + \frac{1}{3}z = A\left(1 - \frac{1}{4}z\right) + B\left(1 - \frac{1}{2}z\right).$$

Evaluate at the poles $z = 2$ and $z = 4$:

$$\begin{aligned} z = 2: \quad 1 + \frac{2}{3} &= A\left(1 - \frac{1}{2}\right) = \frac{1}{2}A \quad \Rightarrow \quad A = 2 \cdot \frac{5}{3} = \frac{10}{3}, \\ z = 4: \quad 1 + \frac{4}{3} &= B\left(1 - 2\right) = -B \quad \Rightarrow \quad B = -\frac{7}{3}. \end{aligned}$$

Each term expands as a geometric series convergent on the unit disk (since $|z| \leq 1 < 2, 4$):

$$\frac{A}{1 - \frac{1}{2}z} = A \sum_{j \geq 0} \left(\frac{1}{2}\right)^j z^j, \quad \frac{B}{1 - \frac{1}{4}z} = B \sum_{j \geq 0} \left(\frac{1}{4}\right)^j z^j.$$

Reading off the coefficient of z^j ,

$$\boxed{\psi_j = \frac{10}{3} \left(\frac{1}{2}\right)^j - \frac{7}{3} \left(\frac{1}{4}\right)^j, \quad j \geq 0}$$

which already gives $\psi_0 = \frac{10}{3} - \frac{7}{3} = 1$. In particular

$$\begin{aligned} \psi_0 &= \frac{10}{3} - \frac{7}{3} = 1, \\ \psi_1 &= \frac{10}{3} \cdot \frac{1}{2} - \frac{7}{3} \cdot \frac{1}{4} = \frac{5}{3} - \frac{7}{12} = \frac{20-7}{12} = \frac{13}{12}, \\ \psi_2 &= \frac{10}{3} \cdot \frac{1}{4} - \frac{7}{3} \cdot \frac{1}{16} = \frac{5}{6} - \frac{7}{48} = \frac{40-7}{48} = \frac{33}{48} = \frac{11}{16}. \end{aligned}$$

Check (coefficient matching). Match powers of z in $(1 - \frac{3}{4}z + \frac{1}{8}z^2) \sum_j \psi_j z^j = 1 + \frac{1}{3}z$:

$$\psi_0 = 1, \quad \psi_1 - \frac{3}{4}\psi_0 = \frac{1}{3} \Rightarrow \psi_1 = \frac{3}{4} + \frac{1}{3} = \frac{13}{12},$$

and $\psi_j = \frac{3}{4}\psi_{j-1} - \frac{1}{8}\psi_{j-2}$ for $j \geq 2$, giving $\psi_2 = \frac{3}{4} \cdot \frac{13}{12} - \frac{1}{8} = \frac{13}{16} - \frac{1}{8} = \frac{11}{16}$, matching exactly.